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§9. Representations of finite groups I

group representation = group action on vector spaces preserving linear structure

nonlinear ↓ group theory ⇝ representation theory ↓ linear

Notation. Let V be a vector space over the complex numbers $\mathbb{C}$.

$GL(V) = \{\mathbb{C}\text{-linear automorphisms } V \xrightarrow{\sim} V\}$

(if $\dim V = n$, choosing $V \simeq \mathbb{C}^n$ get $GL(V) \simeq GL_n(\mathbb{C})$.)

DEF Let G be a group and V a vector space over $\mathbb{C}$.

A (linear) representation of G on V is a homomorphism

$$\rho: G \longrightarrow GL(V)$$
$$g \longmapsto \rho(g): V \xrightarrow{\sim} V$$

ie. $\forall g,h \in G$, $\rho(gh) = \rho(g) \circ \rho(h)$.

(equivalently, a representation is an action $G \curvearrowright V$ that is $\mathbb{C}$-linear

$G \times V \to V$
$(g, v) \mapsto \rho(g)(v)$ $\quad \rho(g): V \xrightarrow{\sim} V$ $\mathbb{C}$-linear)

We write (ρ, V) or just ρ or V for the representation.

Later we'll focus on finite group G and finite-dim vector spaces V

but the def is general.

eg. 1) 1-dim representation $\longleftrightarrow$ $\rho: G \to GL(\mathbb{C}) = \mathbb{C}^\times$ (homomorphism; abelian)

Recall $\mathrm{Hom}(G, \mathbb{C}^\times) \simeq \mathrm{Hom}(G/G', \mathbb{C}^\times)$ (bijection) $\Rightarrow \{1\text{-dim reps of } G\} \longleftrightarrow \{1\text{-dim reps of } G/G'\}$.

2) Set $V := \mathbb{C}[G] = \left\{\sum_{x \in G} a_x x \;\middle|\; a_x \in \mathbb{C},\ a_x = 0 \text{ for all but finitely many}\right\}$

($\mathbb{C}[G]$: vector space with basis G)

$G \curvearrowright V$ $\mathbb{C}$-linearly $\quad \rho(g)\left(\sum_{x \in G} a_x x\right) := \sum_{x \in G} a_x (g \cdot x)$.

This is called the (left) regular rep of G.

3) More generally, if X is a set, $G \curvearrowright X$ get a rep of G on $\mathbb{C}[X]$ = vector space with basis X.

$\rho(g)(\sum a_x x) := \sum a_x (g \cdot x)$.

DEF If (ρ, V) and (ρ', V') are two reps of G,

a $\mathbb{C}$-linear map $\Phi: V \to V'$ is called a homomorphism of representation of G (G-linear for short) if

$\forall g \in G,\ \Phi \circ \rho(g) = \rho'(g) \circ \Phi$ ie. Φ is G-equivariant

If Φ moreover is bijective, then it is an isomorphism of rep of G.

eg. if X and Y are two G-sets and $\psi: X \to Y$ a G-equivariant map,

then ψ induces a G-linear $\Phi: \mathbb{C}[X] \to \mathbb{C}[Y]$, $\sum a_x x \mapsto \sum a_x \psi(x)$

GOAL Given a (finite) group G

Classify all (finite-dim) reps of G up to isoms.

DEF Let (ρ, V) be a rep of G.

A subrepresentation is a $\mathbb{C}$-linear subspace $W \subset V$

that is stable under G, ie. $\forall g \in G,\ \rho(g)(W) \subset W$

$\rightsquigarrow$ get $\rho_W = \rho|_W : G \to GL(W)$ homom. (ρ_W, W) is a rep of G.

eg. Let G be a finite group

Consider $W = \{a(\sum_{x \in G} x) \mid a \in \mathbb{C}\} \subset \mathbb{C}[G]$ ($\dim W = 1$)

$\forall g \in G \quad \rho(g)\left(a\left(\sum_{x \in G} x\right)\right) = a\left(\sum_{x \in G} gx\right) = a\left(\sum_{x \in G} x\right) \in W$

$\rho_W : G \to GL(W) \cong \mathbb{C}^\times$, $g \mapsto 1$ is the trivial rep.

eg. If $\Phi: V \to V'$ is a homom. of reps of G, then $\ker \Phi$ is a subrep of V.

DEF Let $(\rho, V), (\rho', V')$ be two reps of G

Define their direct sum $\rho \oplus \rho' : G \to GL(V \oplus V')$ $\quad \rho \oplus \rho'(g)(v, v') = (\rho(g)(v), \rho'(g)(v'))$

THM Let G be a finite group and (ρ, V) a rep of G

If $W \subset V$ is a subrep then $\exists$ a subspace $W^\perp$ stable under G, st. $V \overset{G\text{-linear}}{\simeq} W \oplus W^\perp$ "线性"的好处

We call $W^\perp$ a complement rep of W in V.

idea Let $(\rho, V), (\rho', V')$ be two reps of a finite G

Given any $\mathbb{C}$-linear map $\phi: V \to V'$

can obtain a G-linear map $\widetilde{\phi}: V \to V'$ by "averaging over G".

$\widetilde{\phi} := \frac{1}{|G|}\sum_{x\in G} \rho'(x)\circ\phi\circ\rho(x)^{-1}: V \to V'$

$\left(\forall g\in G,\ \widetilde{\phi}\circ\rho(g) = \frac{1}{|G|}\sum_{x\in G}\rho'(x)\circ\phi\circ\rho(x)^{-1}\circ\rho(g) = \frac{1}{|G|}\sum_{x\in G}\rho(x)\circ\phi\circ\rho(\underbrace{x^{-1}g}_{y^{-1}}) = \frac{1}{|G|}\sum_{y\in G}\rho'(gy)\circ\phi\circ\rho(y)^{-1} = \rho'(g)\circ\widetilde{\phi}\right)$

! G finite and $\mathrm{char}(\mathbb{C})=0$ both important.

Pf. of THM. Pick any subspace W', st. $V \overset{\mathbb{C}\text{-linear}}{\simeq} W\oplus W'$

Consider the projection $pr: V \overset{\mathbb{C}\text{-linear}}{\to} W$

Apply the above contruction and get $\widetilde{pr} := \frac{1}{|G|}\sum_{x\in G}\rho(x)\circ pr\circ\rho(x)^{-1}: V \to W$.

$\forall w\in W,\ \widetilde{pr}(w) = \frac{1}{|G|}\sum_{x\in G}\rho(x)\circ pr\circ\underbrace{\rho(x)^{-1}(w)}_{\in W} = \frac{1}{|G|}\sum_{x\in G} w = w$.

So $\widetilde{pr}$ is another projection to W. Now define $W^{\perp} = \ker(\widetilde{pr})$ and get $V \overset{G\text{-linear}}{\simeq} W\oplus W^{\perp}$. □

DEF. A rep V of G is called irreducible if the only subreps of V are 0 and V.

COR. If G is finite, then every finite-dim rep V of G is completely reducible.

ie. V is a direct sum of irreducible rep $V \simeq W_1\oplus W_2\oplus\cdots\oplus W_k$ $\quad W_i$ irreducible.

LEM. (Schur's lemma). Let $(\rho,V),(\rho',V')$ be two irreducible finite-dim reps of G, let $\phi: V\to V'$ be G-linear. Then

1). if $(\rho,V),(\rho',V')$ are not isom., we have $\phi=0$.

2). if $(\rho,V)\simeq(\rho',V')$, then $\phi=\lambda\,\mathrm{id}_V$ for some $\lambda\in\mathbb{C}$.

Pf. if $\phi=0$ nothing to prove. Can assume $\phi\neq 0$.

- $\ker\phi\subset V$ is a subrep. But V is irreducible $\Rightarrow \ker\phi = 0$.
- $\mathrm{im}\phi\subset V'$ ——— But V' ———$\Rightarrow \mathrm{im}\phi = V'$.

This already proves 1) if $(\rho,V)\not\simeq(\rho',V')$ then necessarily $\phi=0$.

for 2), $\phi\in GL(V)$ take an eigenvalue $\lambda\in\mathbb{C}$ ← algebraically closed

Consider $\phi' = \phi - \lambda\,\mathrm{id}_V: V\to V'$.

$\forall g\in G,\ \phi'\circ\rho(g) = \rho(g)\circ\phi' \Rightarrow \phi'$ is also G-linear

But $\ker\Phi' \ni v \neq 0 \xrightarrow{V \text{ irred.}} \ker\Phi' = V$, $\Phi' = 0 \Rightarrow \Phi = \lambda \mathrm{id}_V$. $\square$

COR Let G be a finite group.

Then every finite-dim rep V of G admits a unique decomposition (up to isom. and permutations).

$V \simeq W_1^{\oplus m_1} \oplus W_2^{\oplus m_2} \oplus \cdots \oplus W_r^{\oplus m_r}$ with W_i irreducible and pairwise nonisomorphic.

Pf. uniqueness: $W_1^{\oplus m_1} \oplus \cdots \oplus W_r^{\oplus m_r} \xrightarrow{\Phi} \simeq V \simeq U_1^{\oplus n_1} \oplus \cdots \oplus U_s^{\oplus n_s}$

Schur $\Rightarrow \Phi(W_i^{\oplus m_i}) \subset U_j^{\oplus n_j}$ for a unique j with $W_i \simeq U_j$ $m_i \leq n_j$.

Apply Schur again to Φ^{-1}. $\square$

GOAL Given a finite group G

1) Classify all irreducible finite-dim reps of G.

2) Decompose explicitly every finite-dim rep V into direct sum of irreducibles.

Tensor product and dual

DEF If $(\rho_1, V_1), (\rho_2, V_2)$ are two reps of G

Define their tensor product to be $(\rho, V_1 \otimes V_2)$ on which G acts as

$\rho(g)(v \otimes w) = \rho_1(g)(v) \otimes \rho_2(g)(w)$.

If (ρ, V) is a rep of G

Define the dual rep (ρ^*, V^*) $V^* = \mathrm{Hom}_{\mathbb{C}}(V, \mathbb{C})$.

for $v^* \in V^*$, $v \in V$, $\rho^*(g)(v^*)(v) = v^*(\rho(g^{-1})(v))$.

$\langle \cdot, \cdot \rangle : V^* \times V \to \mathbb{C}$ Then $\langle \rho^*(g)(v^*), v \rangle = \langle v^*, \rho(g^{-1})(v) \rangle$.

$\langle v^*, v \rangle = v^*(v)$

Why g^{-1}?

(Need $\rho^*(gh) = \rho^*(g)\rho^*(h)$. $(\rho^*(g)(\underbrace{(\rho^*(h)(v^*))}_{\in V^*})(v) = (\rho^*(h)(v^*))(\rho(g^{-1})(v)) = v^*(\rho(h^{-1})\rho(g^{-1})(v)) = v^*((\rho(gh)^{-1})(v)) = \rho^*(gh)(v^*)(v)$.)

§10. Representations of finite groups II

From now on, we only consider finite-dim reps V of finite groups G.

KEY INPUT characters of a representation.

DEF Let $\rho: G \to GL(V)$ be a rep, $\dim V = n$. (you may fix $V \overset{\Phi}{\simeq} \mathbb{C}^n$ $\rho_\Phi: G \to GL_n(\mathbb{C})$).

Define $\chi_\rho: G \to \mathbb{C}$ called the 特征标 character of the rep.
$\chi_\rho(g) = \mathrm{tr}(\rho(g))$

LEM 1). $\chi_\rho(1) = n$.

2) $\chi_\rho(hgh^{-1}) = \chi_\rho(g)$

3) $\chi_{\rho^*}(g) = \chi_\rho(g^{-1}) = \overline{\chi_\rho(g)}$ (complex conjugation)